Analysis of Building Frames (Using Kani's Method)

Gasper Kani, a German engineer, developed another distribution procedure based on slope deflection equations. This method is very useful for the analysis of multistorey frames. The greatest advantage of this method is, even if a mistake is committed in distribution in one of the cycles, it converges finally to the correct answer. Even today, many practising engineers who are not familiar with computer methods, use Kani's method for the analysis of 3 to 4 storey building frames.

This method is first explained for structures with fixed ends. Then, the modifications to handle simply supported and overhanging ends are discussed. Analysis of symmetric frames making use of the symmetry is also explained, after which an analysis of general frames is taken up.

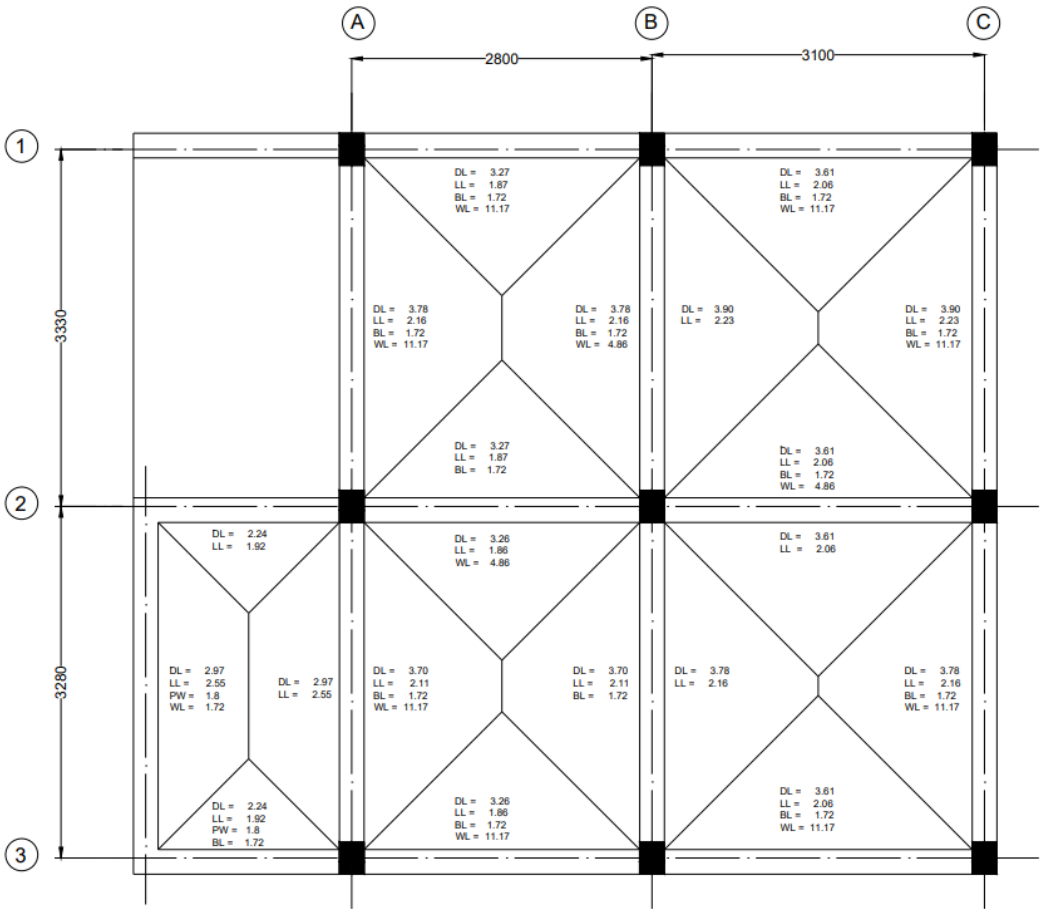


Figure 1: Beam Layout

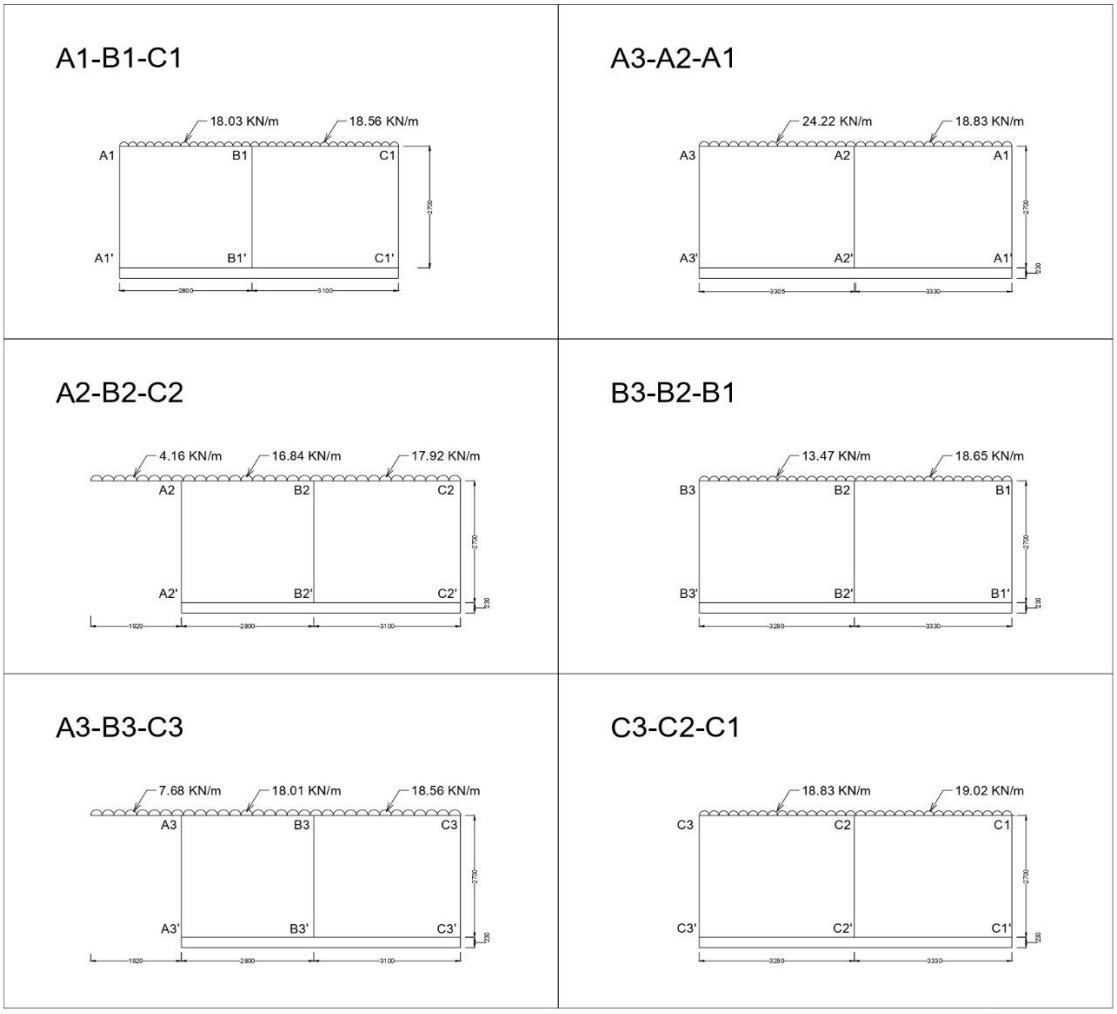
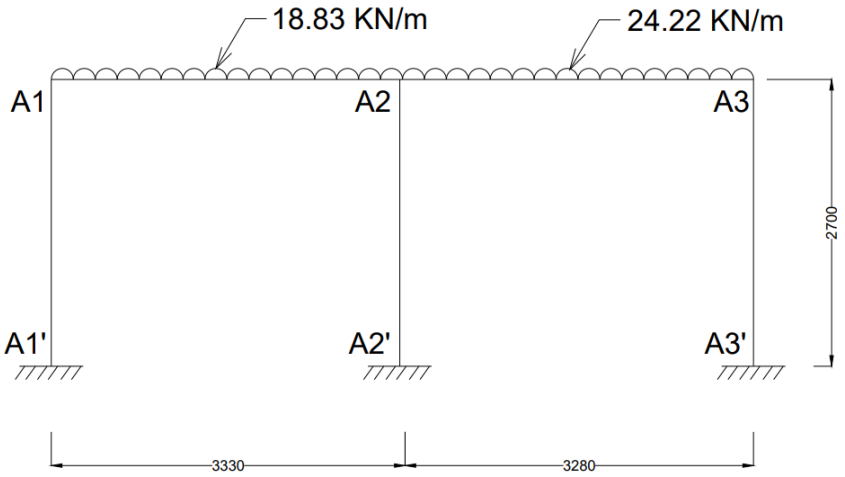


Figure 2: Frames.

Analysis of Frame A1-B1-C1



Step 1: Fixed End Moments

$MF_{A1A2} = \frac{-WL^2}{12} = \frac{-18.83 \times 3.33^2}{12}$ = -17.40 KN-m

$MF_{A2A1} = \frac{WL^2}{12} = \frac{18.83 \times 3.33^2}{12}$ = +17.40 KN-m

$MF_{A2A3} = \frac{-WL^2}{12} = \frac{-24.22 \times 3.28^2}{12}$ = -21.71 KN-m

$MF_{A3A2} = \frac{WL^2}{12} = \frac{24.22 \times 3.28^2}{12}$ = +21.71 KN-m

Fixed end moments in All the columns are zero.

i.e,

$MF_{A1A1'} = MF_{A1'A1} = MF_{A2A2'} = MF_{A2'A2} = MF_{A3A3'} = MF_{A3'A3} = 0$

Step 2: Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right), \left(k = \frac{4EI}{L}\right)$

Joints	Members	Stiffness (k)	Σk	R.F
A1	A1A2	1.201EI	2.682EI	-0.224
	A1A1′	1.481EI		-0.276
A2	A2A1	1.201EI	3.902EI	-0.154
	A2A3	1.220EI		-0.156
	A2A2′	1.481EI		-0.190
A3	A3A2	1.220EI	2.701EI	-0.226
	A3A3′	1.481EI		-0.274

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A1A2} = \left(\frac{4EI}{3.33}\right) = 1.201EI$

$K_{A1A1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{A2A1} = \left(\frac{4EI}{3.33}\right) = 1.201EI$

$K_{A2A3} = \left(\frac{4EI}{3.28}\right) = 1.220EI$

$K_{A2A2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{A3A2} = \left(\frac{4EI}{3.28}\right) = 1.220EI$

$K_{A3A3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$R.F_{A1A2} = -\frac{1}{2}\left(\frac{1.201}{2.682EI}\right) = -0.224$

$R.F_{A1A1'} = -\frac{1}{2}\left(\frac{1.481EI}{2.682EI}\right) = -0.276$

$R.F_{A2A1} = -\frac{1}{2}\left(\frac{1.201EI}{3.902EI}\right) = -0.154$

$R.F_{A2A3} = -\frac{1}{2}\left(\frac{1.220EI}{3.902EI}\right) = -0.156$

$R.F_{A2A2'} = -\frac{1}{2}\left(\frac{1.481EI}{3.902EI}\right) = -0.190$

$R.F_{A3A2} = -\frac{1}{2}\left(\frac{1.220EI}{2.701EI}\right) = -0.226$

$R.F_{A3A3'} = -\frac{1}{2}\left(\frac{1.481EI}{2.701EI}\right) = -0.274$

Sum of Stiffness $\sum k$ at Joints

A1: 1.201EI + 1.481EI = 2.682EI

A2: 1.201EI + 1.220EI + 1.481EI = 3.902EI

A3: 1.220EI + 1.481EI = 2.701EI

Step 3: Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

Storey	Members	k	Σk	D.F
1	A1A1′	1.481EI	4.444EI	-0.5
	A2A2′	1.481EI		-0.5
	A3A3′	1.481EI		-0.5

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A1A1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{A2A2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{A3A3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$D.F_{A1A1'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

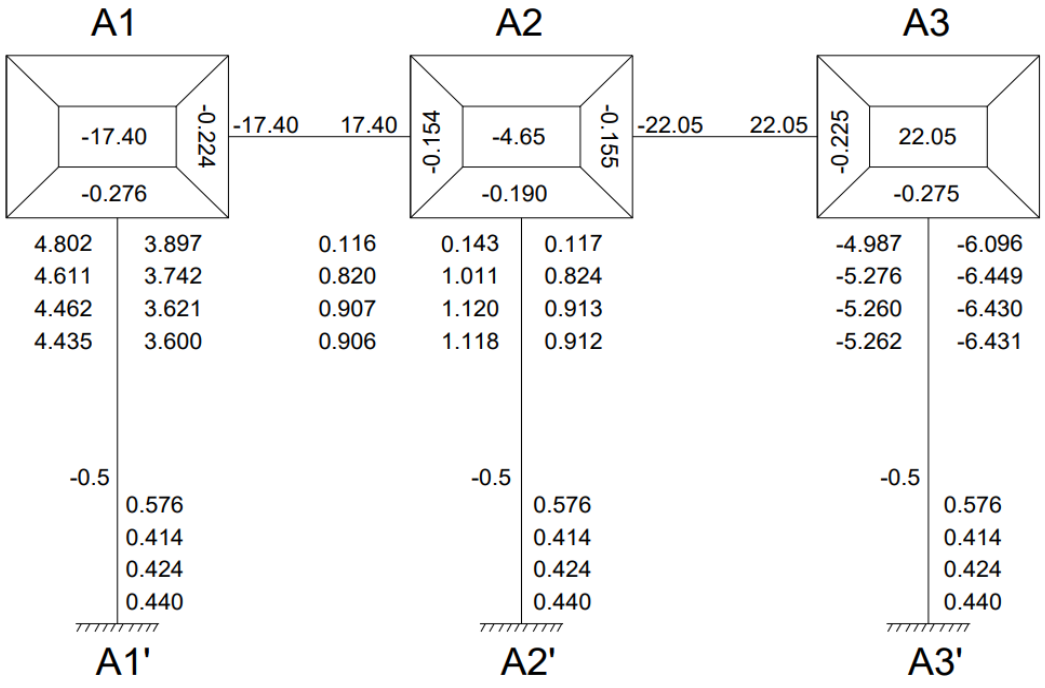
$D.F_{A2A2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{A3A3'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

Sum of Stiffness $\sum k$ at Storey

$\sum k = 1.481EI + 1.481EI + 1.481EI = 4.444EI$

Rotation and Displacement Contribution



Cycle 1

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A1A2} = -0.224 × [-17.40 + 0 + 0] = 3.897

$M_{A1A1'}$ = -0.276 × [-17.40 + 0 + 0] = 4.802

M_{A2A1} = -0.154 × [-4.31 + 3.897 + 0] = 0.064

M_{A2A3} = -0.156 × [-4.31 + 3.897 + 0] = 0.064

$M_{A2A2'}$ = -0.190 × [-4.31 + 3.897 + 0] = 0.078

M_{A3A2} = -0.226 × [21.71+ 0.064 + 0] = -4.920

$M_{A3A3'}$ = -0.274 × [21.71 + 0.064 + 0] = -5.966

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A1A1'} + M_{B1B1'} + M_{C1C1'}$ = -0.5 × (4.802 + 0.078 + (-5.966)) = 0.543

Cycle 2

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A1A2} = -0.224 × [-17.40 + 0.543 + 0.064] = 3.761

$M_{A1A1'}$ = -0.276 × [-17.40 + 0.543 + 0.064] = 4.635

M_{A2A1} = -0.154 × [-4.31 + 0.543 + 3.761 + (-4.920)] = 0.759

M_{A2A3} = -0.156 × [-4.31 + 0.543 + 3.761 + (-4.920)] = 0.768

$M_{A2A2'}$ = -0.190 × [-4.31 + 0.543 + 3.761 + (-4.920)] = 0.936

M_{A3A2} = -0.226 × [21.71 + 0.543 + 0.764] = -5.202

$M_{A3A3'}$ = -0.274 × [21.71 + 0.543 + 0.764] = -6.307

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A1A1'} + M_{B1B1'} + M_{C1C1'}$ = -0.5 × (4.635 + 0.936 + (-6.307)) = 0.368

Cycle 3

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A1A2} = -0.224 × [-17.40 + 0.368 + 0.759] = 3.645

$M_{A1A1'}$ = -0.276 × [-17.40 + 0.368 + 0.759] = 4.491

M_{A2A1} = -0.154 × [-4.31 + 0.368 + 3.645 + (-5.202)] = 0.847

M_{A2A3} = -0.156 × [-4.31 + 0.368 + 3.645 + (-5.202)] = 0.858

$M_{A2A2'}$ = -0.190 × [-4.31 + 0.368 + 3.645 + (-5.202)] = 1.044

M_{A3A2} = -0.226 × [21.71 + 0.368 + 0.858] = -5.183

$M_{A3A3'}$ = -0.274 × [21.71 + 0.368 + 0.858] = -6.285

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A1A1'} + M_{B1B1'} + M_{C1C1'}$ = -0.5 × (4.491 + 1.044 + (-6.285)) = 0.375

Cycle 4

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A1A2} = -0.224 × [-17.40 + 0.375 + 0.847] = 3.624

$M_{A1A1'}$ = -0.276 × [-17.40 + 0.375 + 0.847] = 4.465

M_{A2A1} = -0.154 × [-4.31 + 0.375 + 3.624 + (-5.183)] = 0.846

M_{A2A3} = -0.156 × [-4.31 + 0.375 + 3.624 + (-5.183)] = 0.857

$M_{A2A2'}$ = -0.190 × [-4.31 + 0.375 + 3.624 + (-5.183)] = 1.044

M_{A3A2} = -0.226 × [21.71 + 0.375 + 0.857] = -5.185

$M_{A3A3'}$ = -0.274 × [21.71 + 0.375 + 0.857] = -6.286

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A1A1'} + M_{B1B1'} + M_{C1C1'}$ = -0.5 × (4.435 + 1.118 + (-6.431)) = 0.389

Step 4: Final End Moments

M_{ij} = FEM + 2×M_{ij} (Near end Rotation) + M_{ji} (Far end Rotation) +M″ (Displacement)

Beams (No Sway M″= 0)

M_{A1A2} = -17.40 + 2(3.624) + 0.846 = -9.30

M_{A2A1} = 17.40 + 2(0.846) + 3.624 = 22.72

M_{A2A3} = -21.71 + 2(0.857) + (-5.185) = -25.18

M_{A3A2} = 21.71 + 2(-5.185) + 0.857 = 12.20

Column (With Sway M″= 0.389)

$M_{A1A1'}$ = 0 + 2(4.465) + 0 + 0.389 = 9.32

$M_{A1'A1}$ = 0 + 2(0) + 4.465 + 0.389 = 4.85

$M_{A2A2'}$ = 0 + 2(1.044) + 0 + 0.389 = 2.48

$M_{A2'A2}$ = 0 + 2(0) + 1.044 + 0.389 = 1.43

$M_{A3A3'}$ = 0 + 2(-6.286) + 0 + 0.389 = -12.18

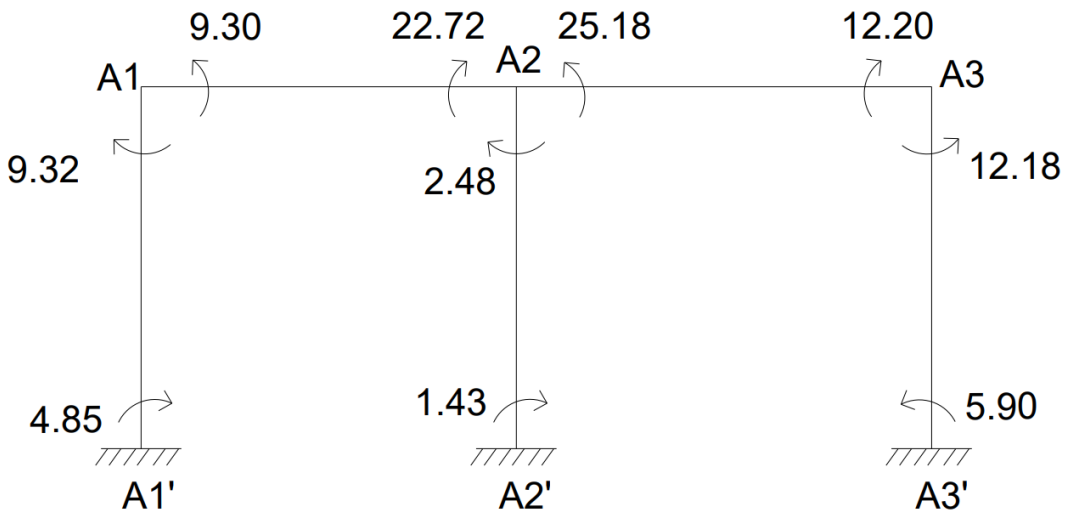
$M_{A3'A3}$ = 0 + 2(0) + (-6.286) + 0.389 = -5.90

Equilibrium Check

Joint A1 = M_{A1A2} + $M_{A1A1'}$ = -9.30 + 9.32 = 0.02

Joint A1 = M_{A2A1} + $M_{A2A3'}$ + $M_{A2A2'}$ = 22.72 + (-25.18) + 2.48 = 0.02

Joint A1 = M_{A3A2} + $M_{A3A3'}$ = 12.20 – 12.18 = 0.02

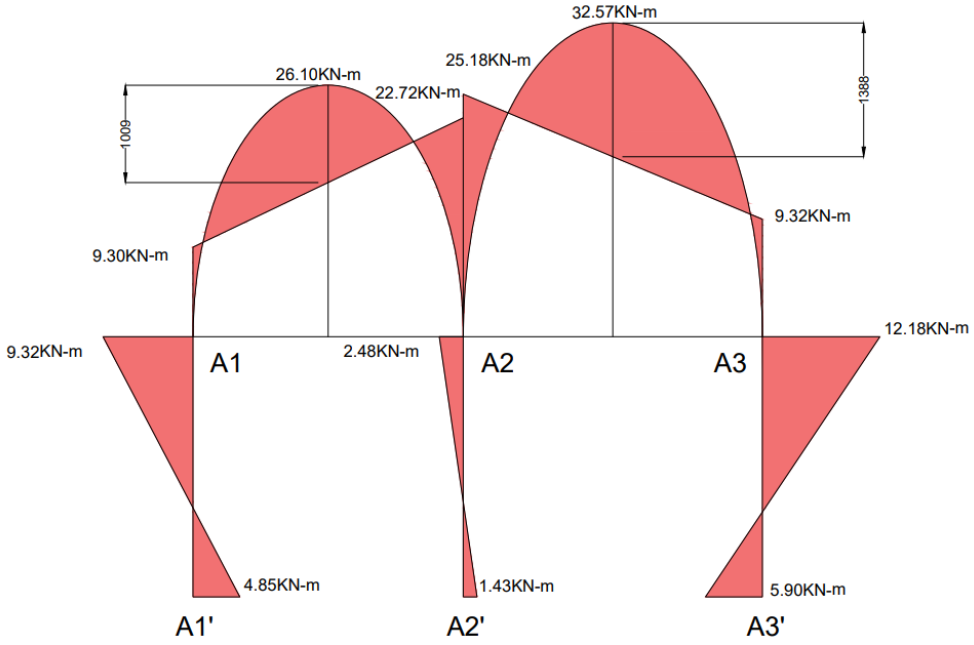


CALCULATION BENDING MOMENT

$BM_{A1A2} = \frac{WL^2}{8} = \frac{18.83 \times 3.33^2}{8} = 26.10KN\text{-}m$

$BM_{A2A3} = \frac{WL^2}{8} = \frac{24.22 \times 3.28^2}{8} = 32.57KN\text{-}m$

BENDING MOMENT DIAGRAM



CALCULATION OF REACTIONS

SPAN A1-A2

Length (L) = 3.33m, UDL (W) = 18.83KN/m

Simply Supported Reactions:

$R_{total} = \frac{WL}{2} = \frac{18.83 \times 3.33}{2} = 31.35KN$

Reaction Due to End Moments:

$R_{moments} = \frac{M_{B1(Left)} - M_{A1)}}{L} = \frac{22.72 - 9.30}{3.33} = 4.03KN$

Final Reactions:

$R_{A1} = R_{total} - R_{moments} = 31.35 - 4.03 = 27.32KN$

$R_{B1(Left)} = R_{total} + R_{moments} = 31.35 + 4.03 = 35.38KN$

SPAN A1-B1

Length (L) = 3.28m, UDL (W) = 24.22KN/m

Simply Supported Reactions:

$R_{total} = \frac{WL}{2} = \frac{24.22 \times 3.28}{2} = 39.72KN$

Reaction Due to End Moments:

$R_{moments} = \frac{M_{B1(Right)} - M_{C1)}}{L} = \frac{25.18 - 8.05}{3.1} = 3.02KN$

Final Reactions:

$R_{B1(Right)} = R_{total} + R_{moments} = 39.72 + 3.02 = 42.74KN$

$R_{C1} = R_{total} - R_{moments} = 39.72 - 3.02 = 36.7KN$

Total Reactions:

Reaction at Column A1 (R_{A1}) = 27.32KN

Reaction at Column B1 (R_{B1}) = $R_{B1(Left)} + R_{B1(Right)} = 35.38 + 42.74 = 78.12KN$

Reaction at Column C1 (R_{C1}) = 36.7KN